Lecture 4: Linear Regression of Time-Series Data I

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Announcement

- In Sam's announcement on UFORA:

  - from 14:00 - 15:45 for Microeconometrics, followed by Time Series from 16:00 - 17:15

- In Timetable:

  - from 14:00 - 16:30 for Microeconometrics, followed by Time Series from 16:30 - 18:30

Linear Regression of Time-Series Data I

Introduction

Recap: Stationary Time-Series Processes

- We have studied MA(q), AR(p), and ARMA(p, q) processes.
- So far:
  - Knowledge: prameters (e.g., ρ, θ, σ_ε^2, etc)
  - Wants: the moments of $\{X_t\}$ (e.g., mean, variance, autocovariance)
- In practice:
  - Knowledge: the moments of $\{X_t\}$ (e.g., mean, variance, autocovariance)
  - Wants: prameters (e.g., ρ, θ, σ_ε^2, etc)
- Q. Can we recover parameters from the moments?

Recap: Stationary Time-Series Processes

- A natural starting point is the linear regression framework.
- It serves as a unified framework for analyzing
  - Intertemporal relationship of a single variable
  - Intratemporal relationship between multiple variables
  - Both inter- and intra-temporal relationships between multiple variables

Recap: Stationary Time-Series Processes

- Intertemporal relationship of a single variable:

Date	Inflation Rate	Unemployment
2024Dec	2.4	6.3
2025Jan	2.5	6.3
2025Feb	2.3	6.3
2025Mar	2.2	6.4
2025Apr	2.2	6.3
2025May	1.9	6.4
2025Jun	2	6.4
2025Jul	2	6.4
2025Aug	2	6.4
2025Sep	2.2	6.4
2025Oct	2.1	6.4
2025Nov	2.1	6.3
2025Dec	2	6.3
2026Jan	1.7	6.2

Recap: Stationary Time-Series Processes

- Moving Average (MA) model of order q:

$$y_t = \gamma_0 + \gamma_1 \varepsilon_{t-1} + \ldots + \gamma_q \varepsilon_{t-q} + \varepsilon_t$$

- Autoregressive (AR) model of order p:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \varepsilon_t$$

- Autoregressive Moving Average (ARMA) model of order p and q:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \varepsilon_t + \gamma_1 \varepsilon_{t-1} + \ldots + \gamma_q \varepsilon_{t-q}$$

Recap: Stationary Time-Series Processes

- Intratemporal relationship between multiple variables:

Date	Inflation Rate	Unemployment
2024Dec	2.4	6.3
2025Jan	2.5	6.3
2025Feb	2.3	6.3
2025Mar	2.2	6.4
2025Apr	2.2	6.3
2025May	1.9	6.4
2025Jun	2	6.4
2025Jul	2	6.4
2025Aug	2	6.4
2025Sep	2.2	6.4
2025Oct	2.1	6.4
2025Nov	2.1	6.3
2025Dec	2	6.3
2026Jan	1.7	6.2

Recap: Stationary Time-Series Processes

- Static model:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + \varepsilon_t$$

Recap: Stationary Time-Series Processes

- Both inter- and intra-temporal relationships between multiple variables:

Date	Inflation Rate	Unemployment
2024Dec	2.4	6.3
2025Jan	2.5	6.3
2025Feb	2.3	6.3
2025Mar	2.2	6.4
2025Apr	2.2	6.3
2025May	1.9	6.4
2025Jun	2	6.4
2025Jul	2	6.4
2025Aug	2	6.4
2025Sep	2.2	6.4
2025Oct	2.1	6.4
2025Nov	2.1	6.3
2025Dec	2	6.3
2026Jan	1.7	6.2

Recap: Stationary Time-Series Processes

- Finite distributed lag (FDL) model of order r:

$$y_t = \beta_0 + \beta_1 z_t + \beta_2 z_{t-1} + \ldots + \beta_{r+1} z_{t-r} + \varepsilon_t$$

- Autoregressive finite distributed lag (AR-FDL) model of order p and r :

$$y_t = \beta_0 + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \gamma_1 z_t + \gamma_2 z_{t-1} + \ldots + \gamma_{r+1} z_{t-r} + \varepsilon_t$$

Recap: Stationary Time-Series Processes

- Time-series model with a time trend:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \gamma t + \varepsilon_t$$

- Time-series model with seasonal components:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \gamma_1 D_{Jan} + \gamma_1 D_{Feb} + \ldots + \gamma_{12} D_{Dec} + \varepsilon_t,$$

whhere D_* is a dammy variable indicating whether time t corresponds to month $*$.

Outline

- Preliminary 1: Math & statistics essentials
- Preliminary 2: Procedure of econometric analysis
- Preliminary 3: Linear regression of cross-sectional data
- Linear regression of time-series data

Linear Regression of Time-Series Data I

Preliminary 1: Math & Statistics Essentials

Math & Statistics Essentials

- Joint probability mass/density function of X and Y:
  - discrete: $P(X = x, Y = y)$
  - continuous: $f(x, y)$

- Joint cumulative distribution function of X and Y:
  - discrete: $P(X \leq x, Y \leq y) = \sum_{u \leq x} \sum_{v \leq y} P(X = u, Y = v)$
  - continuous: $P(X \leq x, Y \leq y) = F(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v) du dv$

- Conditional probability mass/density function of X given $Y = y$:
  - discrete: $P(X = x | Y = y) = \frac{P(X=x, Y=y)}{P(Y=y)}$
  - continuous: $f(x|y) = \frac{f(x,y)}{f(y)}$

- Conditional cumulative distribution function of X given $Y = y$:
  - discrete: $P(X \leq x | Y = y) = \sum_{u \leq x} P(X = u | Y = y)$
  - continuous: $F(x|y) = \int_{-\infty}^{x} f(u|y) du$

Math & Statistics Essentials

- Conditional mean of X given $Y = y$:
  - X is discrete: $E[X|Y = y] = \sum_x x \; P(X = x|Y = y)$
  - X is continuous: $E[X|Y = y] = \int x \; dF(x|y)$

- Conditional expectation is a linear operator: for rv's X, Y, and Z, and a constant $a \in \mathbb{R}$,
  (1) Additivity: $E[X + Y|Z] = E[X|Z] + E[Y|Z]$
  (2) Homogeneity: $E[aX|Z] = aE[X|Z]$

- Useful property:

$$E[XY|Y] = YE[X|Y]$$

- Law of total expectation (a.k.a. Law of iterated expectation):

$$E[X] = E[E[X|Y]]$$

Proof. For the case of discrete random variables,

$$\begin{aligned}
E[E[X|Y]] &= \sum_y E[X|Y]P(Y=y) = \sum_y \left(\sum_x xP(X=x|Y=y) \right) P(Y=y) \\
&= \sum_y \left(\sum_x x \frac{P(X=x, Y=y)}{P(Y=y)} \right) P(Y=y) \\
&= \sum_x x \underbrace{\sum_y P(X=x, Y=y)}_{P(X=x)} = \sum_x xP(X=x) \\
&= E[X].
\end{aligned}$$

The case of continuous random variables is analogous. $\square$

- Conditional variance of X given Y:
  - Definition: $Var(X|Y) = E[(X - E[X|Y])^2|Y]$
  - Implication: $Var(X|Y) = E[X^2|Y] - (E[X|Y])^2$

    Proof.

$$\begin{aligned} Var(X|Y) &= E[(X - E[X|Y])^2|Y] = E[X^2 - 2E[X|Y]X + (E[X|Y])^2|Y] \\ &= E[X^2|Y] - 2\underbrace{E[X|Y]E[X|Y]}_{(E[X|Y])^2} + \underbrace{E[(E[X|Y])^2|Y]}_{(E[X|Y])^2} \\ &= E[X^2|Y] - (E[X|Y])^2 \end{aligned}$$

- Conditional variance is NOT a linear operator.
  - This is left for your exercise.

- Law of Total Variance:

$$Var(X) = E[Var(X|Y)] + Var(E[X|Y])$$

Proof. On the one hand,

$$E[Var(X|Y)] = E[E[X^2|Y] - E[X|Y]^2] = \underbrace{E[E[X^2|Y]]}_{E[X^2]} - E[E[X|Y]^2] = E[X^2] - E[E[X|Y]^2].$$

On the other hand,

$$Var(E[X|Y]) = E[E[X|Y]^2] - (\underbrace{E[E[X|Y]]}_{E[X]})^2 = E[E[X|Y]^2] - (E[X])^2.$$

Adding these two yields

$$E[Var(X|Y)] + Var(E[X|Y]) = E[X^2] - (E[X])^2 = Var(X|Y).$$

- Conditional covariance of X and Y given Z:
  - Definition: $Cov(X, Y|Z) = E[(X - E[X|Z])(Y - E[Y|Z])|Z]$
  - Implication: $Cov(X, Y|Z) = E[XY|Z] - E[X|Z]E[Y|Z]$

    Proof.

$$
\begin{aligned}
Cov(X, Y|Z) &= E[(X - E[X|Z])(Y - E[Y|Z])|Z] \\
&= E[XY - XE[Y|Z] - YE[X|Z] + E[X|Z]E[Y|Z]|Z] \\
&= E[XY|Z] - E[X|Z]E[Y|Z] - E[Y|Z]E[X|Z] + \underbrace{E[E[X|Z]E[Y|Z]|Z]}_{E[X|Z]E[Y|Z]} \\
&= E[XY|Z] - E[X|Z]E[Y|Z]
\end{aligned}
$$

- Conditional covariance is a linear operator.
  - This is left for your exercise.

- Law of total covariance:

$$Cov(X, Y) = E[Cov(X, Y|Z)] + Cov(E[X|Z], E[Y|Z])$$

Proof. On the one hand,

$$\begin{aligned} E[Cov(X, Y|Z)] &= E[E[XY|Z] - E[X|Z]E[Y|Z]] \\ &= \underbrace{E[E[XY|Z]]}_{E[XY]} - E[E[X|Z]E[Y|Z]] \\ &= E[XY] - E[E[X|Z]E[Y|Z]]. \end{aligned}$$

- Law of total covariance:

$$Cov(X,Y) = E[Cov(X,Y)|Z] + Cov(E[X|Z], E[Y|Z])$$

Proof (cont'd). On the other hand.

$$\begin{aligned} Cov(E[X|Z], E[Y|Z]) &= E[E[X|Z]E[Y|Z]] - \underbrace{E[E[X|Z]]}_{E[X]}\underbrace{E[E[Y|Z]]}_{E[Y]} \\ &= E[E[X|Z]E[Y|Z]] - E[X]E[Y]. \end{aligned}$$

Adding these two yields

$$E[Cov(X,Y)|Z] + Cov(E[X|Z], E[Y|Z]) = E[XY] - E[X]E[Y] = Cov(X,Y).$$

□

Linear Regression of Time-Series Data I

Preliminary 2: Big Picture of Econometric Analysis

Procedure of Econometric Analysis

- Econometric analysis proceeds in five steps:

  1. **Definition**

  2. **Data**

  3. **Identification**

  4. **Estimation**

  5. **Inference, Forecasting**

Procedure of Econometric Analysis: Definition

- A researcher comes up with a research question.
- The research question determines **the object of interest $\boldsymbol{\theta}$**.
- This also determines the **hypothesis** to be tested.
- Economists develop an economic model to describe the mechanism behind θ.
- "I feel consumption is more sensitive to an increase in income than widely thought."

  $\rightarrow \theta$: the responsiveness of consumption with respect to income.

  $\rightarrow H_0 : \theta = 1$ vs. $H_0 : \theta > 1$.

Procedure of Econometric Analysis: Data

- Given the research question, the researcher collects data X.
- The observables X are assumed to be a realization from a probability distribution P_X.

Procedure of Econometric Analysis: Identification

- In this stage, we ask if θ is "well-defined" in terms of the observables (X, P_X).
  - qualitative question
- This stage assumes we have full knowledge about the observables (X, P_X).
- We say that θ is **identified** if there exists a unique map between θ and (X, P_X).
- Q. If we knew (X, P_X) pefectly, would it be possible to recover θ from (X, P_X)?
- Q. What else do we need to know?

Procedure of Econometric Analysis: Identification

- Example: An **AR(1)** process (with $|\rho| < 1$):

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t$$

$\boldsymbol{\rho}$:

$$\rho = \frac{Cov(X_t, X_{t-1})}{Var(X_t)}$$

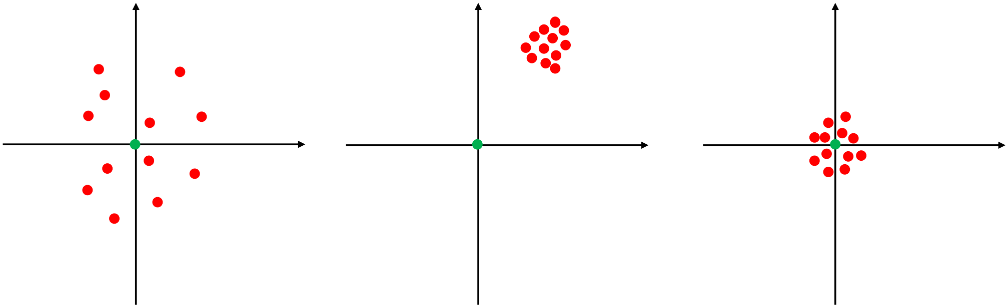
$\hookrightarrow$ ρ is recovered from $\{X_t\}$.

Procedure of Econometric Analysis: Estimation

- In this stage, we approximate θ by the observed variables $\{X_n\}_{n=1}^{N}$.
  - quantitative question
- This stage constructs an estimator $\widehat{\theta}(\{X_n\}_{n=1}^{N})$.
- e.g., $\theta = E[X] = \int x dF(x)$
  $\Longrightarrow$ a sample-analog estimator: $\widehat{\theta}(\{X_n\}_{n=1}^{N}) := \frac{1}{N}\sum_{n=1}^{N} X_n$
- The performance of an estimator is assessed by two concepts:
  1. **Accuracy:** The closeness of a given set of measurements to the true value.
     - Mean: $E[\widehat{\theta}] - \theta$
  2. **Precision:** The closeness of the measurements with each other.
     - Variance: $Var(\widehat{\theta})$

Procedure of Econometric Analysis: Estimation

- Intuition: Darts



accurate, but not precise | not accurate, but precise | accurate and precise

: true value
: an estimate

Procedure of Econometric Analysis: Inference

- This stage tests the hypothesis (H_0 vs. H_1) based on the estimator $\widehat{\theta}$.
- This stage requires the sampling distribution of $\widehat{\theta}$ to construct a test statistic under H_0.

Five Step Procedure of (Two-Sided) Hypothesis Testing

(Step 1): Determine the hypothesis: $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$.

(Step 2): Specify the level of significance α.

(Step 3): Derive the sampling distribution of $\widehat{\theta}$ and construct the test statistic z under H_0.

(Step 4): Compute the critical value $z_{\alpha/2}$: $P(|z| < z_{\alpha/2}) = 1 - \alpha$.

(Step 5): Reject H_0 if and only if $z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$.

Linear Regression of Time-Series Data I

Preliminary 3: Cross-Sectional Linear Regression (Finite-Sample Properties)

Cross-Sectional Linear Regression: Identification

Assumption 1 (CS1): Linearity

A random sample $\{(y_n, x_{n1}, \ldots, x_{nK}) : n = 1, \ldots, N\}$ follows

$$y_n = \beta_1 x_{n1} + \beta_2 x_{n2} + \ldots + \beta_K x_{nK} + \varepsilon_n,$$

where ε_n is an unobserved random error.

- N: sample size, K: number of explanatory variables
- Typically, $x_{n1} = 1$ for all $n \in \{1, \ldots, N\}$.
- In matrix notation, $Y = X\beta + \varepsilon$, where

$$Y = \begin{bmatrix} y_{11} \\ \vdots \\ y_{N1} \end{bmatrix}, X = \begin{bmatrix} x_{11} & x_{12} & \ldots & x_{1K} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \ldots & x_{NK} \end{bmatrix}, \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_K \end{bmatrix}, \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_N \end{bmatrix}.$$

Cross-Sectional Linear Regression: Identification

Assumption 2 (CS2): No perfect multicollinearity

None of the explanatory variables is a perfect linear combination of the others.

- Economics: There is no redundant regressors.

  e.g., $y_n = \beta_1 x_{n1} + \beta_2 x_{n2} + \beta_3 x_{n3} + \varepsilon_n$

  - NG: $x_{n3} = 0.3x_{n1} + 1.2x_{n2}$

  - OK: $x_{n3} \neq ax_{n1} + bx_{n2}$ for any $a, b \in \mathbb{R}$

- Mathematics: The inverse matrix $(X'X)^{-1}$ exists.

Cross-Sectional Linear Regression: Identification

Assumption 3 (CS3): Strict Exogeneity

$E[\varepsilon_n|X] = 0$ for all $n \in \{1, \dots, N\}$.

- No omitted variables nor endogenous variables

  e.g., True model: $y_n = \beta_0 + \beta_1 x_n + \beta_2 x_n^2 + u_n$

  Our model: $y_n = \beta_0 + \beta_1 x_n + \varepsilon_n$

  $\Longrightarrow \varepsilon_n = \beta_2 x_n^2 + u_n$

  $\Longrightarrow$ CS3 fails...

- Under CS3,

  - $E[X'\varepsilon] = E[E[X'\varepsilon|X]] = E[X'E[\varepsilon|X]] = 0$

Cross-Sectional Linear Regression: Identification

Proposition: Identification

Under CS1–CS3, the parameter vector β is identified.

Proof. Under CS1,

$$Y = X\beta + \varepsilon.$$

Premultiplying by X',

$$X'Y = X'X\beta + X'\varepsilon.$$

Take expectation,

$$E[X'Y] = E[X'X]\beta + \underbrace{E[X'\varepsilon]}_{0 \text{ by CS3}}.$$

Under CS2, $(E[X'X])^{-1}$ exists, so that

$$\beta = (E[X'X])^{-1}E[X'Y].$$

This is a unique map between β and (X, Y), establishing the identification of β. $\square$

Cross-Sectional Linear Regression: Estimation

- Given the identification result

$$\beta = (E[X'X])^{-1}E[X'Y],$$

a natural estimator for β is the sample analog:

$$\widehat{\beta} = (X'X)^{-1}(X'Y).$$

- We want to study the stochastic property of $\widehat{\beta}$.
  - Accuracy
  - Precision

Cross-Sectional Linear Regression: Estimation

Proposition: Unbiasedness of $\widehat{\beta}$

Under CS1–CS3, $\widehat{\beta}$ is unbiased for β, i.e., $E[\widehat{\beta}|X] = \beta$.

Proof. By CS1,

$$\widehat{\beta} = (X'X)^{-1}(X'(X\beta + \varepsilon)) = (X'X)^{-1}(X'X)\beta + (X'X)^{-1}(X'\varepsilon) = \beta + (X'X)^{-1}(X'\varepsilon).$$

Take expectation,

$$\begin{aligned} E[\widehat{\beta}|X] &= E[\beta + (X'X)^{-1}(X'\varepsilon)|X] \\ &= \beta + (X'X)^{-1}\underbrace{E[X'\varepsilon|X]}_{=0 \text{ by CS2}} \\ &= \beta. \end{aligned}$$

$\square$

Cross-Sectional Linear Regression: Estimation

Assumption 4-1 (CS4-1): Homoskedasticity

$E[\varepsilon_n^2|X] = \sigma^2 > 0$ for all $n \in \{1, \ldots, N\}$.

Assumption 4-2 (CS4-2): No Correlation

$E[\varepsilon_i \varepsilon_j|X] = 0$ for all $i, j \in \{1, \ldots, N\}$ such that $i \neq j$.

- In matrix notation, CS4-1 and CS4-2 can be jointly written as

$$E[\varepsilon\varepsilon'|X] = \sigma I_N,$$

where I_N is an identitiy matrix.

Cross-Sectional Linear Regression: Estimation

Theorem: Gauss-Markov Theorem

Under CS1–CS4-2, $Var(\widehat{\beta}|X) \geq Var(\widehat{b}|X)$, where $\widehat{b}$ is any other unbiased estimator for β.

- Among linear, unbiased estimators, $\widehat{\beta}$ has the lowest variance (i.e., the most precise).
  - efficient
- $\widehat{\beta}$ is called the **Best Linear Unbiased Estimator** (**BLUE**).
- The sampling variance of $\widehat{\beta}$:
  - if σ^2 is known, $Var(\widehat{\beta}|X) = \sigma^2(X'X)^{-1}$
  - if σ^2 is unknown, $\widehat{Var}(\widehat{\beta}|X) = \widehat{\sigma}^2(X'X)^{-1}$, where $\widehat{\sigma}^2 := \frac{1}{N-K}\widehat{\varepsilon}'\widehat{\varepsilon}$.

Cross-Sectional Linear Regression: Inference

- Let β_k be the kth element of β.
- We want to test

$$H_0 : \beta_k = \bar{\beta}_k$$
$$H_1 : \beta_k \neq \bar{\beta}_k,$$

  where $\bar{\beta}_k$ is the hypothesized value of β_k.

- The level of significance is α.
  - Usually, α ranges from 0.01 to 0.10.

Assumption 5 (CS5): Normality

$\varepsilon_n | X \sim \mathcal{N}(0, \sigma^2)$ i.i.d. for $n \in \{1, \ldots, N\}$.

- ε_n is independently and identically distributed to a normal (a.k.a. Gaussian) distribuion.
- In matrix notation,

$$\varepsilon | X \sim \mathcal{N}(0, \sigma^2 I_N)$$

- This assumption implies CS3, CS4-1 and CS4-2.

Cross-Sectional Linear Regression: Inference

Proposition: Sampling Distribution of $\widehat{\beta}$

Under CS1–CS5,

(i) if σ^2 is known: $z_k|X \sim \mathcal{N}(0,1)$, where $z_k := \frac{\widehat{\beta}_k - \bar{\beta}_k}{(Var(\widehat{\beta}|X)_{kk})^{1/2}}$;

(ii) if σ^2 is unknown: $t_k|X \sim t(N-K)$, where $t_k := \frac{\widehat{\beta}_k - \bar{\beta}_k}{(\widehat{Var}(\widehat{\beta}|X)_{kk})^{1/2}}$.

Proof (Sketch). Under H_0,

$$\widehat{\beta} = \beta + (X'X)^{-1}X'\varepsilon \qquad \Longrightarrow \qquad \frac{\widehat{\beta} - \bar{\beta}}{(Var(\widehat{\beta}|X))^{1/2}}\Bigg| X \sim \mathcal{N}(0, I_K).$$

Picking the kth element, we obtain $z_k|X \sim \mathcal{N}(0,1)$. The case of σ^2 unknown is analogous but technically complicated. □

Cross-Sectional Linear Regression: Inference

Five-Step Procedure of (Two-Sided) Hypothesis Testing

(Step 1): Determine the hypothesis $H_0 : \beta_k = \bar{\beta}_k$ vs. $H_1 : \beta_k \neq \bar{\beta}_k$.

(Step 2): Specify the level of significance α.

(Step 3): Derive the sampling distribution of $\widehat{\beta}_k$ and construct the test statistic under H_0:

- σ^2 is known: $z_k|X \sim \mathcal{N}(0, 1)$;
- σ^2 is unknown: $t_k|X \sim t(N - K)$

(Step 4): Compute the critical value $z_{\alpha/2}$ or $t_{\alpha/2}$:

- σ^2 is known: $P(|z_k| < z_{\alpha/2}) = 1 - \alpha$;
- σ^2 is unknown: $P(|t_k| < t_{\alpha/2}) = 1 - \alpha$

(Step 5): Reject H_0 if and only if

- σ^2 is known: $z_k^* < -z_{\alpha/2}$ or $z_k^* > z_{\alpha/2}$;
- σ^2 is unknown: $t_k^* < -t_{\alpha/2}$ or $t_k^* > t_{\alpha/2}$

Linear Regression of Time-Series Data I

Finite-Sample Properties

Time-Series Linear Regression: Identification

Assumption 1 (TS1): Linearity

The stochastic process $\{y_t, x_{t1}, \dots, x_{tK}\}_{t=1}^{T}$ follows

$$y_t = \beta_1 x_{t1} + \beta_2 x_{t2} + \ldots + \beta_K x_{tK} + \varepsilon_t,$$

where $\{\varepsilon_t\}_{t=1}^{T}$ is the sequence of unobserved errors.

- y_t, x_t and ε_t are not assumed to be strongly nor weakly statioinary.
- In matrix notation, $Y = X\beta + \varepsilon$, where

$$Y = \begin{bmatrix} y_1 \\ \vdots \\ y_T \end{bmatrix}, X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1K} \\ \vdots & \vdots & \ddots & \vdots \\ x_{T1} & x_{T2} & \dots & x_{TK} \end{bmatrix}, \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_T \end{bmatrix}, \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_T \end{bmatrix}.$$

Time-Series Linear Regression: Identification

Assumption 2 (TS2): No Perfect Collinearity

None of the explanatory variables is a perfect linear combination of the others.

- This is the same as CS2.

Time-Series Linear Regression: Identification

Assumption 3 (TS3): Strict Exogeneity

$E[\varepsilon_t|X] = 0$ for all $t \in \{1, \ldots, T\}$.

- The expected value of ε_t, given the explanatory variables for all time periods, is zero.
- Under TS3,
  - $E[\varepsilon_t] = E[E[\varepsilon_t|X]] = 0$
  - $E[x_{sk}\varepsilon_t] = E[E[x_{sk}\varepsilon_t|X]] = E[x_{sk}E[\varepsilon_t|X]] = 0$ for all $s \in \{1, \ldots, T\}$ and $k \in \{1, \ldots, K\}$

Time-Series Linear Regression: Identification

- Three messages implied by $E[x_{sk}\varepsilon_t] = 0$:
  - $s < t$: ε_t is uncorrelated with past x_{sk}.
  - $s = t$: ε_t is uncorrelated with contemporaneous x_{sk}.
  - $s > t$: ε_t is uncorrelated with future x_{sk}.

- Ex) an AR(1) model
  - $y_t = \alpha + \rho y_{t-1} + \varepsilon_t$
  - $y_{t+1} = \alpha + \rho \underbrace{y_t}_{x_{t+1}} + \varepsilon_{t+1}$

    $\Longrightarrow$ when $s = t + 1$, TS3 fails...

- TS3 excludes any model with lagged dependent variables.
  - AR(p), ARMA(p,q), AR-FDL models, etc.

Time-Series Linear Regression: Identification

Proposition: Identification

Under TS1–TS3, the parameter vector β is identified.

- This is the same as the cross-sectional case.
- β is identified as

$$\beta = (E[X'X])^{-1}E[X'y].$$

Time-Series Linear Regression: Estimation

- Identified parameter: $\beta = (E[X'X])^{-1}E[X'y]$
- Estimator: $\widehat{\beta} = (X'X)^{-1}X'Y$

Proposition: Unbiasedness

Under TS1–TS3, $\widehat{\beta}$ is unbiased for β, i.e., $E[\widehat{\beta}|X] = \beta$.

Time-Series Linear Regression: Estimation

Assumption 4-1 (TS4-1): Homoskedasticity

$E[\varepsilon_t^2|X] = \sigma^2$ for all $t \in \{1, \ldots, T\}$.

Assumption 4-2 (TS4-2): No Serial Correlation

$E[\varepsilon_s \varepsilon_t|X] = 0$ for all $s, t \in \{1, \ldots, T\}$

- Under TS3 and TS4-1,
  - $Var(\varepsilon_t|X) = E[\varepsilon_t^2|X] - (E[\varepsilon_t|X])^2 = \sigma^2$
  - $Var(\varepsilon_t) = E[Var(\varepsilon_t|X)] - Var(E[\varepsilon_t|X]) = \sigma^2$
- Under TS3 and TS4-2,
  - $Cov(\varepsilon_s, \varepsilon_t|X) = E[\varepsilon_s \varepsilon_t|X] - E[\varepsilon_s|X]E[\varepsilon_t|X] = 0$
  - $Cov(\varepsilon_s, \varepsilon_t) = E[Cov(\varepsilon_s, \varepsilon_t|X)] - Cov(E[\varepsilon_s|X], E[\varepsilon_t|X]) = 0$
- Under TS3, TS4-1, and TS4-2, $\varepsilon_t \sim WN(0, \sigma^2)$.
  - $\{\varepsilon_t\}$ is weakly stationary, but not necessarily strongly stationary.

Time-Series Linear Regression: Estimation

- TS4-2 excludes any model with an autoregressive error term.
- E.g., a static model with an AR(1) error term:

$$y_t = \alpha + \beta x_t + \gamma e_{t-1} + e_t,$$

where $e_s \sim WN(0, \sigma_e^2)$ for all $s \leq t$.

This can be written as

$$y_t = \alpha + \beta x_t + \varepsilon_t,$$

where $\varepsilon_t := \gamma e_{t-1} + e_t$.

However,

$$E[\varepsilon_t \varepsilon_{t-1} | X] = \gamma \sigma_e^2 > 0,$$

violating TS4-2.

Time-Series Linear Regression: Estimation

Theorem: Gauss-Markov Theorem

Under TS1–TS4-2, $Var(\widehat{\beta}|X) \geq Var(\widehat{b}|X)$, where $\widehat{b}$ is any other unbiased estiamtor for β.

- Among linear, unbiased estimators, $\widehat{\beta}$ has the lowest variance (i.e., the most precise).
  - efficient
- $\widehat{\beta}$ is called the **Best Linear Unbiased Estimator** (**BLUE**).
- The sampling variance of $\widehat{\beta}$:
  - if σ^2 is known, $Var(\widehat{\beta}|X) = \sigma^2(X'X)^{-1}$
  - if σ^2 is unknown, $\widehat{Var}(\widehat{\beta}|X) = \widehat{\sigma}^2(X'X)^{-1}$, where $\widehat{\sigma}^2 := \frac{1}{N-K}\widehat{\varepsilon}'\widehat{\varepsilon}$.

Time-Series Linear Regression: Inference

Assumption 5 (TS5): Normality

$\varepsilon_t | X \sim \mathcal{N}(0, \sigma^2)$ i.i.d. for $t \in \{1, \ldots, T\}$.

- ε_n is independently and identically distributed to a normal (a.k.a. Gaussian) distribuion.
- In matrix notation,

$$\varepsilon | X \sim \mathcal{N}(0, \sigma^2 I_N)$$

- This assumption implies TS3, TS4-1 and TS4-2.
- Under (TS3, TS4-1, TS4-2, and) TS5, $\varepsilon_t \sim GWN(0, \sigma^2)$.
  - $\{\varepsilon_t\}$ is both weakly and strongly stationary.

Time-Series Linear Regression: Inference

Proposition: Inference

Under TS1–TS5, the five-step procedure for hypothesis testing is valid.

Time-Series Linear Regression: Inference

- Examples of time-series models satisfying TS1–TS5:
  - Ex1) Static model:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + \varepsilon_t$$

  - Ex2) Time-series model with a time trend:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \gamma t + \varepsilon_t$$

  - Ex3) Time-series model with seasonal components:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \gamma_1 D_{Jan} + \gamma_1 D_{Feb} + \ldots + \gamma_{12} D_{Dec} + \varepsilon_t,$$

  - Ex4) Finite distributed lag (FDL) model of order r:

$$y_t = \beta_0 + \beta_1 z_t + \beta_2 z_{t-1} + \ldots + \beta_{r+1} z_{t-r} + \varepsilon_t$$

Time-Series Linear Regression: Inference

- Examples of time-series models **NOT** satisfying TS1–TS5:
  - Ex1) Autoregressive (AR) model of order p:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \varepsilon_t$$

$\hookrightarrow$ TS3 fails...

  - Ex2) Autoregressive finite distributed lag (AR-FDL) model of order p and r :

$$y_t = \beta_0 + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \gamma_1 z_t + \gamma_2 z_{t-1} + \ldots + \gamma_{r+1} z_{t-r} + \varepsilon_t$$

$\hookrightarrow$ TS3 fails...

  - Ex3) Autoregressive Moving Average (ARMA) model of order p and q:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \varepsilon_t + \gamma_1 \varepsilon_{t-1} + \ldots + \gamma_q \varepsilon_{t-q}$$

$\hookrightarrow$ TS3 and TS4-2 fail...

Time-Series Linear Regression: Model Selection

- Q. Which model should be chosen?
  - e.g., $y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$ or $y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + \varepsilon_t$?
- Akaike Information Criterion (AIC):

$$AIC := 2ln(L) - 2K,$$

  where
  - L: the likelihood function evaluated at the observed values of X and Y
  - K: the number of parameters
- Choose the model that maximizes the AIC.

Summary and Preview

Summary

- Econometric analysis of time-series data
  - (definition, data,) identification, estimation, and inference
- We apply the classical linear regression analysis to time-series data.
  - finite-sample properties
- $\{\varepsilon_t\}$ are WN or GWN.
  - No stationarity assumption on $\{x_t\}$ and $\{y_t\}$
- Theoretical properties remain the same.
  - accuracy, precision

Preview

- TS3 excludes any model with lagged dependent variables.
  - e.g., AR, AR-FDL, ARMA models
- TS4-2 excludes any model with an autoregressive error term.
  - e.g., ARMA models
- Next, we will consider large-sample properties.